

Supplementary Examinations - May - June 2023

20EC501 – ANALOG COMMUNICATION

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I

Marks BT* CO* PO*

- | | | | | | | |
|----|-------|---|---|-----|---|---|
| 1. | a) | An honest wheel of chance is divided into three equal segments coloured green (G), red (R), and yellow (Y). The wheel is spun twice and the outcome in the form of resulting colour sequence GR, RG, ... etc., is taken. If A is the event "neither colour is yellow" and B is the event "Matching colours", calculate P(A), P(B), P(AB), and P(A+B). | 8 | L*3 | 1 | 1 |
| | b) | The input to the binary communication system, denoted by a random variable x, takes on one of the two values 0 and 1 with probabilities $\frac{3}{4}$ and $\frac{1}{4}$ respectively. Due to errors cause by the noise in the system the output y differs from the input x occasionally. The behavior of the communication system is modeled by the conditional probabilities $P(y=1/x=1) = \frac{3}{4}$ and $P(y=0/x=0) = \frac{7}{8}$. | | | | |
| | (i) | Find P(y=1) and P(y=0) | | | | |
| | (ii) | Find P(x=1/y=1) | 6 | L3 | 1 | 1 |
| | c) | Consider the probability density function $f_X(x) = ke^{- x }$ for $-\infty < x < \infty$. Find | | | | |
| | (i) | The value of k | | | | |
| | (ii) | $F_X(x)$ | 6 | L3 | 1 | 1 |
| 2. | a) | Consider a random process $X(t) = \cos(2\pi f_c t + \Theta)$, where A and f_c are constants and Θ is a uniformly distributed random variable over the interval $(-\pi, \pi)$. Find | | | | |
| | (i) | The autocorrelation function of X(t) | | | | |
| | (ii) | Power spectral density of X(t) | | | | |
| | (iii) | Average power of X(t) | 8 | L3 | 1 | 1 |
| | b) | Two random variables X and Y are related by the expression $Y = aX + b$, where a and b are real numbers. | | | | |
| | (i) | Show that their correlation coefficient is | | | | |
| | | $\rho = \begin{cases} 1, & \text{if } a > 0 \text{ for any } b \\ -1, & \text{if } a < 0 \text{ for any } b \end{cases}$ | | | | |
| | (ii) | Show that their covariance is | | | | |
| | | $COV[XY] = a\sigma_X^2$ | | | | |
| | | where, σ_X^2 is the variance of X. | 8 | L3 | 1 | 1 |
| | c) | Define power spectral density and list out its properties. | 4 | L2 | 1 | 1 |

Unit – II

3. a) The output current of 60 % modulated AM generator is 1.5 A. To what value will this current rise if the generator is modulated additionally by another audio wave, whose modulation index is 0.7? What will be the percentage power saving if the carrier and one of the sidebands are now suppressed?

- b) Explain the generation of standard AM waves using the square law modulator. Give time- and frequency domain representations of the modulated signal and plot its amplitude spectrum. Also give specifications of the filter used. 10 ;L2 2 1
- c) For the AM signal $s(t) = A_c \cos [2\pi f_c t + \phi(t)]m(t)$, find pre-envelope, complex envelope, natural envelope, and inphase and quadrature components. 4 L3 2 1
4. a) With neat diagrams for the transmitter and receiver, explain the working of quadrature carrier multiplexing system. 8 L2 2 1
- b) With suitable mathematical analysis, obtain an expression for the time-domain representation of the lower-sideband SSB-SC signal using sideband filtering technique. 9 L2 2 1
- c) An AM broadcast station has modulation index which is 0.75 on the average. What would be its average power saving if it could over to single sideband suppressed carrier transmissions, while having to maintain the same signal strength in its reception area? 3 L3 2 1

Unit – III

5. a) In an FM system, when the audio frequency (AF) is 500 Hz and modulating voltage 2.5 V, the deviation produced is 5 kHz. If the modulating voltage is now increased to 7.5 V, calculate the new value of frequency deviation produced. If the AF voltage raised to 10 V while the modulating frequency dropped to 250 Hz, what is the frequency deviation? Calculate the modulation index in each case. 6 L3 3 1
- b) With suitable mathematical analysis, explain narrow-band frequency modulation. List out your observations from the generated amplitude spectrum. 10 L2 3 1
- c) The equation for an FM wave is

$$s(t) = 10 \sin [5.7 \times 10^8 t + 5 \sin (12 \times 10^3 t)]$$
 Calculate; carrier frequency, modulating frequency, modulation index, and frequency deviation. 4 L3 3 1
6. a) Briefly discuss the basic methods for generating frequency modulated signals. 4 L2 3 1
- b) Explain the direct method of FM generation using suitable circuit diagram. Obtain the expression for the instantaneous frequency of the signal generated. 8 L2 3 1
- c) An angle modulated signal is represented by

$$s(t) = 10 \cos [2\pi \times 10^6 t + 5 \sin(2000\pi t) + 10 \sin(3000\pi t)]$$
 volts
 Find i) the power in the modulating signal, ii) the frequency deviation, iii) the deviation ratio, iv) the phase deviation, and v) the approximate transmission bandwidth. 8 L3 3 1

Unit – IV

7. a) Briefly discuss shot noise and thermal noise. 4 L2 4 1
- b) Two resistors 28 k Ω and 51 k Ω are at room temperature $T = 290^\circ$ kelvin. Calculate for a bandwidth of 100 kHz, the mean square value and RMS value of thermal noise voltage; i) for each resistor, ii) for two resistors in series, and iii) for two resistors in parallel. 9 L3 4 1
- c) Determine the noise power at the output of the LPF for the scheme shown in Fig. Q.7.c when the input is white noise with power spectral density $\frac{\eta}{2}$.



Fig. Q.7.c

7 L2 4 1

- | | | | | | |
|-------|--|---|----|---|---|
| 8. a) | Obtain an expression for autocorrelation function of white noise and sketch PSD and autocorrelation function. | 6 | L2 | 4 | 1 |
| b) | Derive an expression for equivalent noise temperature of a single two port network. | 8 | L2 | 4 | 1 |
| c) | A white gaussian noise $\omega(t)$ having zero mean and PSD $\frac{N_0}{2}$ is applied to an ideal low pass filter having bandwidth B_0 & pass band amplitude 1. Determine & sketch the PSD & autocorrelation function of the filter output $n(t)$. | 6 | L2 | 4 | 1 |

Unit – V

- | | | | | | |
|--------|---|----|----|---|---|
| 9. a) | With relevant mathematical analysis, discuss noise in DSB-SC receivers. Obtain expressions for the receiver output and figure of merit. | 10 | L2 | 5 | 1 |
| b) | With suitable mathematical analysis, discuss noise in AM receivers. Obtain expressions for the receiver output and figure of merit. | 10 | L2 | 5 | 1 |
| 10. a) | The PSD of noise at the front-end of the receiver is 0.5×10^{-3} Watts/Hz. The modulating wave $m(t)$ is sinusoidal, with a carrier power of 80 kW, and a sideband power of 10 kW per sideband. The message bandwidth is 5 kHz. Assume the use of an envelope detector in the receiver, determine the output SNR of the system. By how many decibels is this system inferior to a DSB-SC system? | 6 | L3 | 5 | 1 |
| b) | An AM receiver operating with a sinusoidal modulating signal has the following specifications;
$\mu = 0.8$ $[SNR]_0 = 30 \text{ dB}$ What is the corresponding carrier-to-noise ratio? | 4 | L3 | 5 | 1 |
| c) | With suitable mathematical analysis, discuss noise in SSB-SC receivers. Obtain expressions for the receiver output and figure of merit. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
